

**WRITING YOUR THESIS WITH LATEX WITH A
VERY, VERY, VERY LONG TITLE**

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**WRITING YOUR THESIS WITH LATEX WITH A
VERY, VERY, VERY LONG TITLE**

by

JOSEPH OOI & LIM LIAN TZE

**Thesis submitted in fulfilment of the requirements
for the degree of
Doctor of Philosophy**

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ACKNOWLEDGEMENT

Many thanks to Prof. Donald Knuth for giving us $\text{\TeX}$, and Leslie Lamport for $\text{\LaTeX}$.

I first learned $\text{\LaTeX}$ as an undergraduate student in Computer Science at the University of Warwick. Back in Malaysia, I picked it up again while doing my M.Sc. at USM, as part of some productive procrastination (there I've admitted it!!). This coincided with Dr. Dhanesh's and Dr. Azman's efforts in raising awareness about $\text{\LaTeX}$ at USM during NaCSPC'05 – somehow one thing led to another, and I now conduct trainings and consultations on $\text{\LaTeX}$. ☺

Since then, many friends and fellow $\text{\LaTeX}$ users have given feedback and helped relayed important updates from IPS to me, to help improve the class and template. It has actually come to a point where you are too numerous to name! Thank you all, as well as everyone who has attended my talks and workshops, used my various templates, downloaded examples from my website (<http://liantze.penguinattack.org/latextypesetting.html>).

Hope everyone graduates quickly then!

Joseph Ooi: Many thanks to LianTze Lim and wnarffin for maintaining the GitHub repo which contains the $\text{\LaTeX}$ template, without which my attempt at reproducing the likeness of the latest thesis template version from USM would be unachievable. Here's to hoping more people taste the forbidden fruit that is $\text{\LaTeX}$ and forsake Microsoft Office Word. Happy $\text{\LaTeX}$ -ing everyone!

Joseph Ooi: By the way, don't worry about if warnings that the hyper reference to some of your defined acronyms not being defined pop up. They should go away once you use those acronyms via `\ac{...}` or `\acs{...}` or `\acf{...}` or something else at least once in your text. Or you can just ignore them.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF PLATES

LIST OF SYMBOLS

$\lim$	limit
θ	angle in radians

LIST OF ABBREVIATIONS

IPS	Institut Pengajian Siswazah
PPSK	Pusat Pengajian Sains Komputer
USM	Universiti Sains Malaysia
UTMK	Unit Terjemahan Melalui Komputer
SYDUUCF	Something You Don't Usually Use Capitalization For

LIST OF APPENDICES

PENULISAN TESIS DENGAN LATEX

ABSTRAK

Ini merupakan abstrak Melayu untuk tesis USM. Ianya disediakan dengan sistem penyediaan dokumen $\text{\LaTeX}$.

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ABSTRACT

This is the English abstract of a USM thesis. It was prepared with the $\text{\LaTeX}$ document typesetting system.

CHAPTER 1

INTRODUCTION: THE ECHONET LITE - AN OVERVIEW

1.1 Overview

Smart-grid development increasingly depends on communication and control at the consumer side of the electricity system. Demand-Side Management (DSM) coordinates electricity consumption with system conditions through activities such as load scheduling, peak reduction, and consumer feedback. A review of DSM research identifies optimization and coordinated load management as important components of smart-grid operation (**sarker2021progress**). At the residential level, these activities are implemented through Home Energy Management Systems (HEMS), which coordinate appliances, meters, and other connected devices.

The resulting communication is not only between a utility and a customer. It also includes local machine-to-machine exchanges between a HEMS controller and appliances that can report measurements or receive control requests. Devices compliant with ECHONET Lite are relevant to this study because the protocol is designed for communication among home-network devices and has been used in smart-home systems and appliance-control research (**tanakaEvaluationSecureEndtoend2018**).

The security of this communication is consequential because an appliance command can change a physical operating state. An attacker who can observe or inject local traffic may be able to alter the behaviour that the HEMS is intended to coordinate. This chapter consequently narrows the broad smart-grid context to the application-layer security of ECHONET Lite. The discussion establishes the protocol context, identifies the security problem, defines the research objectives and scope, and explains the organization of the thesis.

1.2 ECHONET Lite Protocol: An Overview

ECHONET Lite is an application-layer communication protocol for networked home appliances. The ECHONET Lite specification defines a message format containing an ECHONET Lite header and ECHONET Lite data. The data identifies the source and destination objects and carries service, property, and value information for an operation (**echonet_lite_part2**).

The protocol supports operations such as reading and writing device properties and reporting device state. These operations allow a HEMS controller to request an appliance value, change a property, or receive an autonomous notification. The protocol's structured object and property model supports interoperability across appliance categories, while research on Matter and ECHONET Lite demonstrates the continuing importance of interoperability between smart-home ecosystems (**phamMatterECHONETLite2024**).

The specification describes the communication middleware and frame fields, but it does not provide the application-layer authentication and integrity mechanism investigated in this thesis (**echonet_lite_part2**). Consequently, a receiver cannot rely on the standard frame alone to establish that a command originated from an authorized peer or that its contents were not modified in transit.

1.3 Smart-Home Networks and Cybersecurity Threats

The local home network is an important trust boundary for ECHONET Lite communication. The protocol can be used within a smart-home environment, but network reachability does not by itself prove that a sender is authorized. This distinction is particularly important when a HEMS controls appliances remotely or when a local network includes devices with different security capabilities. A secure end-to-end remote-control system for ECHONET Lite appliances has been investigated in previous work, illustrating that remote access and appliance control require explicit security design (**tanakaEvaluationSecureEndtoend2018**).

This thesis considers three threat classes at the application layer. A replay attack reuses a previously valid command. An injection attack creates or modifies a command so that a receiver performs an unauthorized operation. A denial-of-service attack attempts to prevent legitimate communication by consuming processing, communication, or device resources. These threats are analysed as security requirements for the proposed mechanism; this chapter does not claim a measured prevalence of any threat without a supporting empirical source.

1.4 Research Motivation

The motivation for this research is the gap between the lightweight appliance communication model of ECHONET Lite and the security assurance required when that communication is exposed to an untrusted or remotely reachable network. An application-layer mechanism can make the security decision at the point where the ECHONET Lite message is parsed and interpreted.

The proposed direction is a frame-level extension that authenticates relevant message data, detects modification, and rejects commands that fail freshness or replay validation. Its usefulness must be assessed not only by security behaviour but also by the communication and processing overhead imposed on constrained appliance nodes.

1.5 Problem Statement

This study addresses the following problem: standard ECHONET Lite communication provides an application-level data model for home appliances, but the standard frame does not by itself provide the message authentication, integrity verification, and anti-replay controls required for communication across a potentially untrusted network (**`echonet_lite_part2`**). Without a mechanism that combines origin verification, message integrity, and freshness validation, an attacker who gains access to the communication path may attempt unauthorized command injection or replay.

1.6 Research Objectives

The objectives of this study are:

1. To propose a frame-level security mechanism for ECHONET Lite that provides message authentication and data integrity without changing the protocol's core object and property model.
2. To propose an anti-replay and temporal-validation mechanism within the security extension to reduce the risk of unauthorized command injection and duplicated valid messages.
3. To evaluate the proposed mechanism in terms of security behaviour and performance overhead, including its effect on latency and operational efficiency relative to standard ECHONET Lite communication.

The exact authentication-field size, algorithms, implementation platform, and evaluation metrics are specified in the methodology and implementation chapters after the requirements and design have been established.

1.7 Scope and Limitations

The study is limited to application-layer protection for ECHONET Lite messages. It focuses on message authentication, data integrity, freshness or replay validation, and the processing overhead associated with the proposed extension. It does not claim to solve every security problem in a smart grid or smart home. Physical-layer attacks, radio interference, appliance hardware tampering, general-purpose public-key infrastructure, and the security of unrelated smart-home protocols are outside the primary scope unless required for comparison.

The evaluation results will be bounded by the selected simulation, emulation, or hardware environment. Results obtained from that environment should therefore be interpreted as evidence about the tested implementation and workload, rather than as universal performance values for every ECHONET Lite appliance.

1.8 Research Contribution

The intended contribution is an application-layer security extension for ECHONET Lite that connects protocol-level message processing with authentication, integrity, and replay validation requirements. A second contribution is an implementation and evaluation procedure that makes the overhead of the extension observable. Contribution claims will be finalized only after the design and evaluation results have been reported; design features or performance values are not presented here as empirical findings.

1.9 Research Steps

The research follows five stages. First, the ECHONET Lite message model and relevant threats are analysed to derive security and performance requirements. Second, the authentication and freshness mechanism is designed. Third, the mechanism is implemented and integrated with a standard ECHONET Lite communication path. Fourth, legitimate and adversarial message exchanges are tested in the selected evaluation environment. Finally, the results are analysed using the defined security and performance measures.

1.10 Thesis Organization

Chapter 2 reviews ECHONET Lite, smart-home communication, and relevant security mechanisms. Chapter 3 presents the requirements and design of the proposed extension. Chapter 4 describes its implementation and test environment. The discussion chapter interprets the evaluation results and considers limitations, while the conclusion summarizes the findings and identifies possible future work.

CHAPTER 2

CITATIONS AND BIBLIOGRAPHY

Echonet lite does not have native security measures due to this it relies on the security of the underlying network. Security measures such as encryption and authentication are not built into the protocol itself, whereas are delegated to lower layers of the network stack, starting from layer 4 to layer 2 entirely depending upon RFC 5191, DTLS, or IPsec. **echonet_lite_part2**

This chapter should have been a survey on the history of $\text{\TeX}$ and $\text{\LaTeX}$, and a comparison to conventional word processors in preparing academic documents. Due to lack of time on the author's part, and also the abundance of such discussions on the web, we look at ways to prepare the bibliography and citations instead.

2.1 The *.bib File

First of all, bear in mind that your bibliography file (*.bib files) is like a database. That means you can maintain a centralised list, and reuse it for all your publications. $\text{\LaTeX}$ will only list sources that you actually cite in the text for each document, according to the bibliography and citation style you select in each document. But you can still hack it so that your own publications are listed, even if you did not cite it.

```
@BOOK{latex:companion,  
  title = {The {\LaTeX} Companion},  
  publisher = {Addison-Wesley},  
  year = {2004},  
  author = {Frank Mittelbach and Michel Goossens and  
    Johannes Braams and David Carlisle and Chris  
    Rowley},  
  series = {Addison-Wesley Series on Tools and  
    Techniques for Computer Typesetting},  
  address = {Boston, MA, USA},  
  edition = {2nd}  
}
```

Figure 2.1: A BibTeX Entry

As an example, in `mybib.bib` I created a Bib_{TEX} entry with JabRef, the source text of which is shown in Figure 2.1.

One thing to note about authors' names: Bib_{TEX} recognises “Mittelbach” as the last name for both `Frank Mittelbach` and `Mittelbach, Frank`. So for a name like “Lim Lian Tze”, you would have to specify it as either `Lian Tze Lim` or `Lim, Lian Tze` for Bib_{TEX} to recognise “Lim” as the last name correctly. In addition, if the surname or family name of an author consists of multiple words, enclose it with braces to avoid surprises, like so: `Syed Muhammad Naquib {al-Attas}`.

2.2 Citations using the `natbib` package

The `usmthesis` package imports the `natbib` and `apacite` package which provides flexible citation mechanisms, so see its documentation for more details. On a MiK_{TEX} installation, use the command prompt to issue `methelp --view natbib` and `methelp --view apacite` to access the documentation. On TeXLive, simply type `texdoc natbib` and `texdoc apacite` and the documentation will be displayed automatically, if it's found on your machine.

The basic citation commands are `\citet` and `\citep`, which stands for *textual* and *parenthetical* citation respectively. They take extra arguments, too, for adding notes in the citations.

2.2.1 Author-Year System

The default bibliography style is APA:

- `\citet{latex:companion}` → **latex:companion**<empty citation>
- `\citet[chap.~2]{latex:companion}` → **latex:companion**<empty citation>
- `\citep{latex:companion}` → **(latex:companion)**
- `\citep[chap.~2]{latex:companion}` → **(latex:companion)**
- `\citep[see also][]{latex:companion}` → **(latex:companion)**

- `\citep[see also][chap.~2]{latex:companion} → (latex:companion)`
- `\citet{latex:companion,roberts} → latex:companion; roberts<empty citation>`
- `\citep{latex:companion,roberts} → (latex:companion; roberts)`

You may also want to write only the author’s name or year occasionally:

- `\citeauthor{latex:companion} → latex:companion`
- `\citeyear{latex:companion} → latex:companion`
- `\citeyearpar{latex:companion} → (latex:companion)`

2.2.2 Numeric System

If you prefer the plain, numerical system, do the following steps first:

1. In `usmthesis.cls`, search for the line `\RequirePackage[natbibapa]{apacite}` and modify it to:

```
\RequirePackage[numbers]{natbib}
```

2. In `usmthesis.tex`:

- modify the bibliography styles to:

```
\bibliographystyle{plainnat}
```

```
\bibliographystyleown{plainnat}
```

or any other number system style that you prefer.

You will then get the following citation outputs:

- `\citet{latex:companion} → Mittelbach et al. [1]`
- `\citet[chap.~2]{latex:companion} → Mittelbach et al. [1, chap. 2]`
- `\citep{latex:companion} → [1]`
- `\citep[chap.~2]{latex:companion} → [1, chap. 2]`
- `\citep[see also][]{latex:companion} → [see also 1]`
- `\citep[see also][chap.~2]{latex:companion} → [see also 1, chap. 2]`
- `\citet{latex:companion,roberts} → Mittelbach et al. [1], Roberts [3]`
- `\citep{latex:companion,roberts} → [1, 3]`

- `\citeauthor{latex:companion}` → Mittelbach et al.
- `\citeyear{latex:companion}` → 2004
- `\citeyearpar{latex:companion}` → [2004]

CHAPTER 3

FIGURES, TABLES, EQUATIONS, ALGORITHMS, ETC

(This is supposed to be the design or methodology chapter. Instead, we include examples on inserting figures, tables, mathematical equations... i.e. things that you might want to include in your thesis.)

3.1 Inserting Figures

You can draw diagrams with special $\text{\LaTeX}$ commands, but this may take some extra time to learn. I've had some forays into the `pgf` and `tikz` packages and must say I quite like the results; but as I said, they take time to learn. If you want a faster solution, you can draw your diagrams using other applications, and saving them as graphic files (EPS, PNG, JPG, PDF).

$\text{\LaTeX}$ requires EPS (encapsulated postscript) graphic files when generating DVI output, and PNG, JPG or PDF when generating PDF output.

For exporting to EPS, try <http://www.cloudconvert.com>. It's like a Swiss knife for converting from almost any format, to almost any format.

Do note that IPS **discourages** the use of colours in your thesis, including diagrams and figures. Photographs and colour plates are exceptions to this rule: see Section 3.3.

Here's how to insert a picture with the filename `pythag.eps` or `pythag.png`. I'm going to display it here with 5cm width, and the caption "Pythagoras' Theorem".

```
\begin{figure}[hbt!]\centering
\includegraphics[width=50mm]{pythag}
\caption{Pythagoras' Theorem}\label{fig:pythagoras}
\end{figure}
```

Figure 3.1: Including a Graphics File

The result would be:

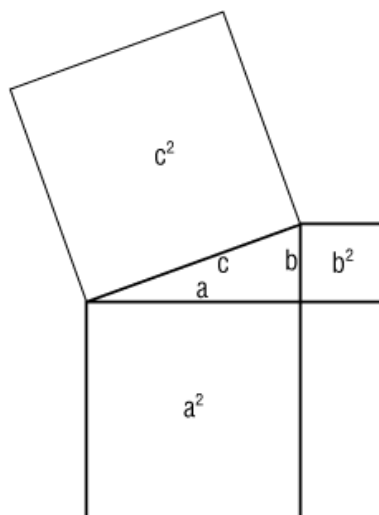


Figure 3.2: Pythagoras' Theroem

Don't specify the extension of the graphic file. The template will automatically look for the EPS or the PNG (or otherwise) versions, depending on whether latex or pdflatex was used. The figure environment will also ensure that that an entry is inserted into the *List of Figures* automatically – including the figure numbering, caption and page number.

In addition, the width of the included graphics can also be specified as a percentage of the text width, e.g. `width=.2\textwidth` would cause the graphics to occupy 20% of the text width.

Notice that I inserted a `\label` just after the `\caption`. This can be used for referencing the figure number, like this:

Figure `\ref{fig:pythagoras}` → Figure 3.2

This works the same for chapters, sections, tables, equations too. In `chap-intro.tex`, I labelled the Introduction chapter with `\label{chap:intro}`. I also labelled the section on inserting figures, `\label{sec:figure}`. So now I can do

Chapter `\ref{chap:intro}` → Chapter 1

section `\ref{sec:figure}` → section 3.1

Everytime the numbering of the heading changes, the reference will change automatically as well. **This is another advantage of using L^AT_EX**: you do not need to manually update the reference counters (nor the Table of Contents, List of Figures and Tables) whenever you add or remove figures, tables, sections or chapters.

You might also want to try out JpgfDraw: it is a vector graphics and drawing application (requiring Java), and can export to L^AT_EX code which you can paste into your L^AT_EX source. JpgfDraw is available from <http://theoval.cmp.uea.ac.uk/~nlct/jpgfdraw/index.html>.

3.2 How Do I Do Subfigures?

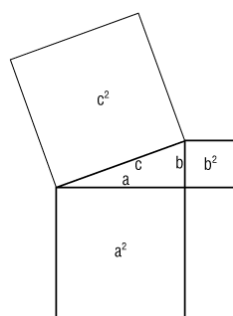
Here's an example on how to do subfigures (and similarly subtables):

```
\begin{figure}[hbt!]
  \begin{minipage}{.49\textwidth}
    \centering
    \subfloat[First caption]{\includegraphics[width=3cm]{
      pythag}} \label{fig:sub1}
  \end{minipage}
  \hfill
  \begin{minipage}{.49\textwidth}
    \subfloat[Second caption]{\includegraphics[width=0.8\textwidth]{USMScience}}\label{fig:sub2}
  \end{minipage}

  \caption{This is the main caption of the figure.}
  \label{fig:main}
\end{figure}
```

Figure 3.3: Creating subfigures within figures.

Joseph Ooi: This is how you can refer to Figure 3.4 and Figure 3.4(a) or even ?? in Figure 3.4 for example. You must make sure to keep your `\label{...}` for the subfigures inside the group, `{...}`, that contains `\includegraphics[]{}{}`. If you don't, the hyperlinks will show up as ??, as was the case in the original code. This has been fixed.



(a) First caption



(b) Second caption

Figure 3.4: This is the main caption of the figure.

3.3 Inserting Plates

Colour photographs are now regarded as *plates*. They must be listed in the *List of Plates* instead of the List of Figures, and should be printed in colour on glossy photo paper (**ips:thesis:guideline:2007**).

The usmthesis document class defines a new plate environment, as well as a corresponding `\listofplates` command. (The `\listofplates` command is already placed in the sample template file `usmthesis.tex`.) In short, all you need to do to insert a photograph or plate (as a graphics file `USMSscience.{eps,png,jpg}`) is shown in Figure 3.5, and you will then get Plate 3.1 as the result.

```
\begin{plate}[hbt!]\centering
\includegraphics[width=.9\textwidth]{USMSscience}
\caption{School of Computer Sciences, USM}\label{plate:
  ppsk:usm}
\end{plate}
```

Figure 3.5: Inserting a Plate

3.4 Inserting Tables

Typesetting tables can be a little troublesome especially with complex layouts. Look up (**roberts**) to learn about some tips, or you can use the LaTeX program (<http://www.g32.org/latable/>) to help you.



Plate 3.1: School of Computer Sciences, USM

If using LaTeX, when you're done designing the table, copy the whole table as $\text{\LaTeX}$ code, and paste it in your source file. (You may add additional formatting commands, like bold, italics, etc.) If this is going to be a numbered table, remember to surround it with `\begin{table}` and `\end{table}`, and give it a caption, like this:

```
\begin{table}[hbt!]\centering
\begin{tabular}{|l|c||r|}
\hline
\textbf{Name} & \textbf{Category} & \textbf{Quantity} \\
\hline
\hline
Apple & Fruit & 10 \\
\hline
Cucumber & Vegetable & 25 \\
\hline
Daisy & Flower & 5 \\
\hline
\end{tabular}
\caption{Sample Table Only} \label{table:sample}
\end{table}
```

Figure 3.6: Typesetting Tables

Table 3.1: Sample Table Only

Name	Category	Quantity
Apple	Fruit	10
Cucumber	Vegetable	25
Daisy	Flower	5

Note also that `usmthesis` is configured such that captions for figures are placed *below* the figures, and captions for tables are placed *above* them, in accordance with the formatting guidelines.

Many of us would have had massive headaches about lining up decimal places in table columns (as mentioned in the IPS guidelines) if not for this tip from (**latex:companion**). This method uses the `dcolumn` package (already loaded by `usmthesis.cls`). Instead of using `l`, `c` or `r` as the column type in the tabular declaration, use

`D{input sep}{output sep}{decimal places}`.

```
\begin{table}[htb!]\centering
\begin{tabular}{| c | D{.}{.}{2} |}
\hline
Item & \multicolumn{1}{c|}{Reading}\\\hline
A & 1.11\\\hline
B & 3.99\\\hline
C & 2.27\\\hline
\end{tabular}
\caption{A table with decimal data}
\end{table}
```

Figure 3.7: Aligning decimal data in tables

The $\text{\LaTeX}$ code in Figure 3.7 will give you Table 3.2.

Table 3.2: A table with decimal data

Item	Reading
A	1.11
B	3.999
C	22.2

Without using `dcolumn`, you'd get something like this:

Table 3.3: A table with decimal data (mis-aligned)

Item	Reading
A	1.11
B	3.999
C	22.2

3.5 Full-paged, Sideways Figures and Tables

To make a figure appear on a landscape, full-page layout, put your `\includegraphics` command in a `sidewaysfigure` environment (Figure 3.8).

```
\begin{sidewaysfigure}\centering
\includegraphics[width=\textheight]{latex-win-comp}
\caption{A full-page, sideways figure}\label{fig:
sidewaysfig}
\end{sidewaysfigure}
```

Figure 3.8: Including a sideways, full-page graphic

The resultant figure (Figure 3.9) should appear on the next page.

For a sideways table, use the `sidewaystable` environment instead around your usual tabular material.

3.6 Mathematical Equations

Typesetting mathematical material is one of, if not *the*, strongest capabilities of $\text{\LaTeX}$. After all, that was the Knuth’s main motivation for creating $\text{\TeX}$. As it is impossible to enumerate all possible mathematically-related commands and macros here, we will just give some examples. The reader is directed to the many well-written online tutorials, such as (**roberts**), for more elaborate examples. $\text{\TeX}$ nicCenter also provides many shortcut buttons for inserting mathematical symbols.

$$z^2 = x^2 + y^2 \tag{3.1}$$

$$\phi = \frac{1}{2}(1 + \sqrt{5}) \tag{3.2}$$

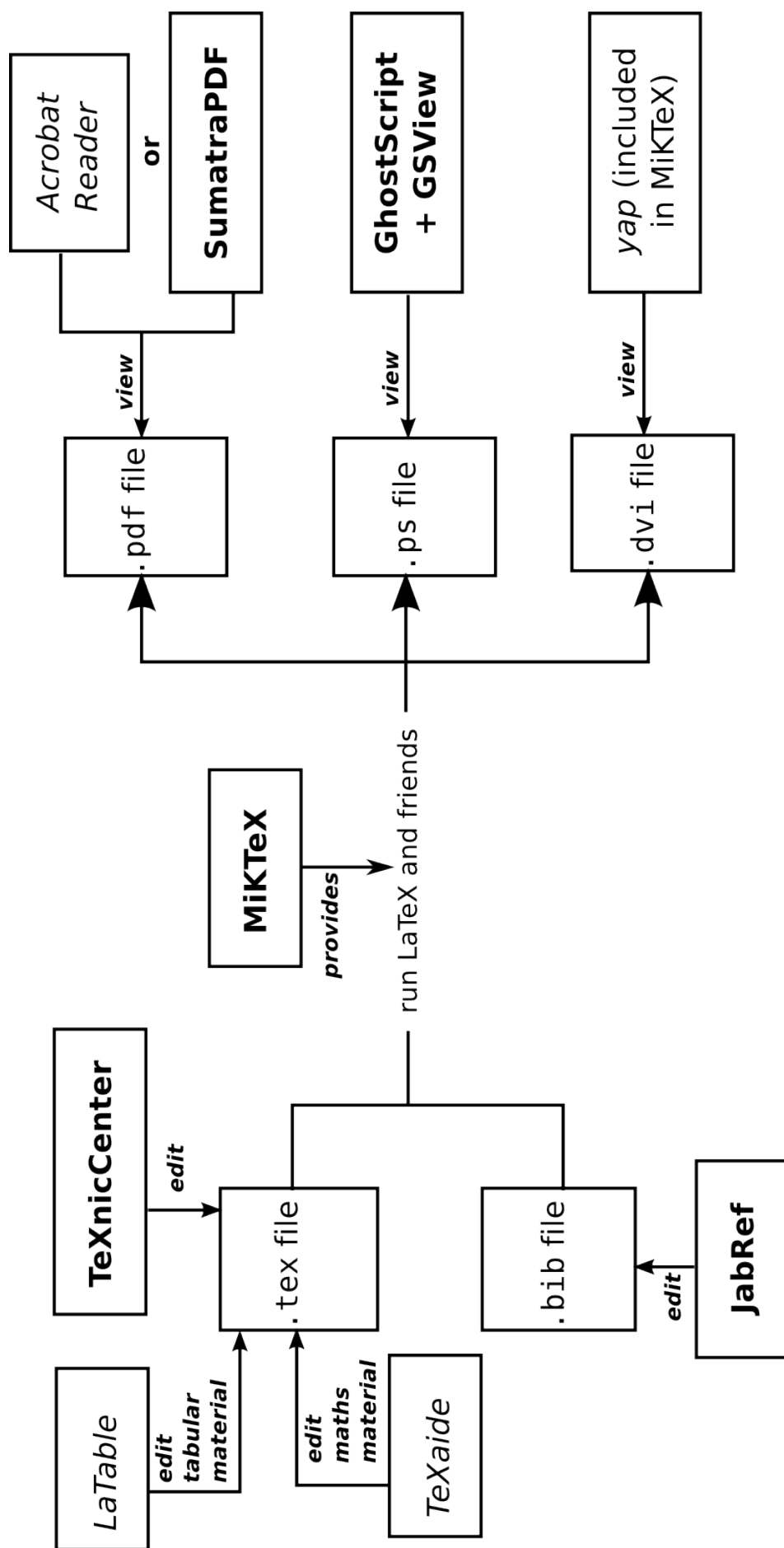


Figure 3.9: A full-page, sideways figure

```

\begin{equation}\label{eq:pythagoras}
z^2 = x^2 + y^2
\end{equation}

\begin{equation}\label{eq:golden:ratio}
\phi = \frac{1}{2} (1 + \sqrt{5})
\end{equation}

\begin{equation}\label{eq:golden:ratio}
\phi = \frac{1}{2} (1 + \sqrt{5})
\end{equation}
\begin{equation}\label{eq:golden:ratio:fibonacci}
\phi = 1 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{F_n F_{n+1}}
\end{equation}

Equation~\ref{eq:pythagoras} is the Pythagoras Theorem.
\eqref{eq:golden:ratio} gives the golden ratio  $\phi$ ,
and
\eqref{eq:golden:ratio:fibonacci} relates it to the
Fibonacci
series.

```

Figure 3.10: Typesetting Mathematical Equations

$$\phi = 1 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{F_n F_{n+1}} \quad (3.3)$$

Equation 3.1 is the Pythagoras Theorem. (3.2) gives the golden ratio ϕ , and (3.3) relates it to the Fibonacci series.

The $\text{\LaTeX}$ code to generate the above mathematics materials are shown in Figure 3.10. As you can see, references to equations can be achieved with either `\ref` or `\eqref`.

A disclaimer: if you think the mathematic equations don't look as great as all those $\text{\LaTeX}$ advocates make them out to be, that's because IPS requires Times to be used and the current offerings of free $\text{\LaTeX}$ math fonts for Times don't look great. It would've been a different picture if we used Computer Modern.

3.7 Acronyms

If you have a list of acronyms or symbols, edit the file `loa.tex` as in Figure 3.11.

```
\begin{acronym}[UTMK] %% replace 'UTMK' with the
    longest acronym in your list
\acro{IPS}{Institut Pengajian Siswazah}
\acro{PPSK}{Pusat Pengajian Sains Komputer}
\acro{USM}{Universiti Sains Malaysia}
\acro{UTMK}{Unit Terjemahan Melalui Komputer}
\end{acronym}
```

Figure 3.11: The template `loa.tex` for acronyms

You can also use this acronym list to help expand it the first time you mention it in your text. For example, the first time you use `\ac{USM}`, ‘Universiti Sains Malaysia (USM)’ will be the output (without the quotes). After that, all calls to `\ac{USM}` will give ‘USM’ (without the quotes). For more information, see the documentation for the acronym package.

3.8 Typesetting Algorithms

As computer scientists, it is quite common to include algorithms and/or pseudocode. There are a number of different packages available, but unfortunately they tend not to work well together! I’m using `algorithmicx` here.

Algorithm 1 Computing $x^n, n > 0$

Require: $n \geq 0$

Ensure: $y = x^n$

$y \leftarrow 1$

$X \leftarrow x$

$N \leftarrow n$

while $N \neq 0$ **do**

if N is even **then**

$X \leftarrow X \times X$

$N \leftarrow \frac{N}{2}$

▷ This is a comment

else if N is odd **then**

$y \leftarrow y \times X$

$N \leftarrow N - 1$

end if

end while

```

\begin{algorithm}[hbt!]
\begin{algorithmic}
\Require $n \geq 0$
\Ensure $y = x^n$
\State $y \leftarrow 1$
\State $X \leftarrow x$
\State $N \leftarrow n$
\While{$N \neq 0$}
\If{$N$ is even}
\State $X \leftarrow X \times X$
\State $N \leftarrow \frac{N}{2}$ \Comment{This is
a comment}
\ElseIf{$N$ is odd}
\State $y \leftarrow y \times X$
\State $N \leftarrow N - 1$
\EndIf
\EndWhile
\end{algorithmic}
\caption{Computing $x^n$, $n > 0$}
\end{algorithm}

```

Figure 3.12: Typesetting Algorithms

3.9 Program Listings

You may have noticed that I used the `lstlisting` environment to typeset some of the L^AT_EX examples – with pretty-printing¹, too, including automatic line-breaking. For more information, see the documentation for the `listings` package: it’s available online at <https://ctan.org/pkg/listings?lang=en>.

Just to give some simple example here. For example, to typeset a “Hello World” Java program with syntax highlighting, you can use the following code:

If you want to turn off the syntax highlighting, set `language={}`. (See the `listings` documentation for a list of programming languages for which syntax highlighting is supported.) You can also change the `basicstyle` value to get different effects: e.g. a different font family, size or text formatting.

Here’s another example for a C program:

¹Whether you agree that it *is* pretty is another story altogether.

```

\lstset{basicstyle=\small\ttfamily, language=Java,
        breaklines=true, columns=fullflexible, tabsize=2}
\begin{lstlisting}
public class HelloWorld {
    public static void main( String arg[] ) {
        for (int i = 0; i < 10; i++) {
            System.out.println( "Hello
                                World!" + i);
        }
    }
}
\end{lstlisting}

```

Figure 3.13: Typesetting a Java program listing

```

public class HelloWorld {
    public static void main( String arg[] ) {
        for (int i = 0; i < 10; i++) {
            System.out.println( "Hello World!" + i);
        }
    }
}

```

Figure 3.14: A pretty-printed Java program listing with syntax highlighting

```

\lstset{basicstyle=\sffamily, language=C, breaklines=
        true, columns=fullflexible, tabsize=2}
\begin{lstlisting}
int main() {
    int c = 0;
    c = c + 1;
    printf( "%d", c );
    return 0;
}
\end{lstlisting}

```

Figure 3.15: Typesetting a C program listing

And here is the same C program listing *without* syntax highlighting (by setting `language={}`):

```
int main() {  
    int c = 0;  
    c = c + 1;  
    printf( "%d", c );  
    return 0;  
}
```

Figure 3.16: A pretty-printed C program listing with syntax highlighting

```
int main() {  
    int c = 0;  
    c = c + 1;  
    printf( "%d", c );  
    return 0;  
}
```

Figure 3.17: A C program listing without syntax highlighting

CHAPTER 4

IMPLEMENTATION

Now is the time to “implement” your thesis with $\text{\LaTeX}$. Go forth and typeset!
Happy $\text{\LaTeX}$ ing! ☺

4.1 Printing Your Thesis

This is *very* important. Assuming you’re printing your thesis from Acrobat Reader, make sure the following settings are chosen correctly in the Print window:

- A4 paper size is selected.
- Make sure your Printer settings is using A4 too.
- No page scaling.

Otherwise, the margins of your printed outputs may go horribly wrong. Print one or two pages first to make sure everything looks fine before printing your entire thesis.

CHAPTER 5

DISCUSSION

Just a placeholder for the discussion chapter.

CHAPTER 6

CONCLUSION

T-that's all folks. Have fun with L^AT_EX!

APPENDICES

APPENDIX A DATA USED

Put some test data here.

Joseph Ooi: You can still make sections and subsections as usual. This is essentially the same as you would do for chapters, but instead you **MUST USE** `\appchapter{...}`, **NOT** `\chapter{...}`. The rest are unchanged.

A.1 Example Appendix Section

A.1.1 Example Appendix Subsection

APPENDIX B UML DIAGRAMS

Yet another dummy placeholder for appendix material.

Joseph Ooi: Another reminder to **USE** \appchapter{...} for the appendix chapters, **DO NOT USE** \chapter{...}.